

Development and validation of an analytical method for determining triflumezopyrim residues in multiple paddy field matrices (Paddy Water, Paddy Soil, Brown Rice, Rice Husk, and Rice Straw) using GC-ECD and GC-MS

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panelXinhong Deng <sup>a</sup>, Wenxing Wang <sup>a</sup>, Bin Yang <sup>a</sup>, Xuguo Zhou <sup>b</sup>, Lihua Yang <sup>a</sup>

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Highlights

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Development and validation of novel GC-ECD and GC-MS methods for triflumezopyrim residue analysis in paddy matrices.

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The pretreatment and instrumental conditions have undergone systematic optimization studies.

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The validated GC-ECD and GC-MS methods gave low LOD and satisfactory recoveries.

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Application of the proposed methods to the analysis of real paddy samples from Hunan province in China.

Abstract

Triflumezopyrim (TFM), a neonicotinoid insecticide for rice planthopper control, poses potential risks to rice safety and paddy ecosystems via residues. Herein, gas chromatography-electron capture detector (GC-ECD) and GC-mass spectrometry (GC-MS, selected ion monitoring:  $m/z$  206.0, 116.0, 131.0) methods were developed and validated for TFM residue analysis in paddy matrices. Samples were extracted with dichloromethane and purified by dispersive solid-phase extraction. For five matrices, calibration curves showed good linearity ( $0.01\text{--}1\text{ mg L}^{-1}$ ,  $R^2 \geq 0.9912$ ). At 0.01, 0.1, 1  $\text{mg kg}^{-1}$  spiking levels,

mean recoveries were 81.04%-109.66% (GC-MS) and 89.45%-106.41% (GC-ECD), with relative standard deviations (RSDs) of 3.55%-13.08% and 0.59%-9.30%, respectively. Limits of quantitation (LOQ) were 0.01 mg kg<sup>-1</sup> for both techniques; limits of detection (LOD) were 0.001 mg L<sup>-1</sup> (GC-ECD) and 0.0005 mg L<sup>-1</sup> (GC-MS). These parameters met current residue limits and analytical criteria. Robustness tests revealed minimal impacts of pretreatment and most chromatographic parameters, except carrier gas flow rate (critical, RSD > 10% when varied). The validated method was applied to real paddy field samples from Hunan Province, China, and 20 batches of five types of substrates were screened, with trace TFM in soil (below LOQ). This is the first validated GC-ECD/GC-MS method for TFM trace analysis covering the entire paddy ecosystem (paddy water, paddy soil, brown rice, rice husk, and rice straw), suitable for routine monitoring and supporting TFM environmental/food safety evaluation.

## Introduction

Rice (*Oryza sativa* L.) is a staple food for over half the global population, contributing 56 % to total cereal calorie intake, but it is highly susceptible to key planthoppers (brown, white-backed, and small brown planthoppers) [1,2]. These pests thrive in the warm, humid conditions of rice fields, and severe infestations can cause 30–50 % yield losses [3,4]; chemical control remains the primary management method in conventional agriculture.

Triflumezopyrim (TFM), a novel neonicotinoid insecticide with the chemical name 3,4-dihydro-2,4-dimethyl-1-(pyridin-5-ylmethyl)-3- (3-trifluoromethylphenyl)- 2H-pyridin-1-ium-3-ide, was developed by DuPont Crop Protection [2,5]. This insecticide exhibits high efficacy in controlling planthoppers, primarily by inducing lethargy in the brown planthopper. Notably, it demonstrates strong selectivity toward target pests while exerting minimal adverse effects on non-target organisms, including natural predators such as *Pardosa pseudoannulata* [6]. In contrast to conventional neonicotinoids, which function as activators of the nicotinic acetylcholine receptor (nAChR), TFM acts as an nAChR antagonist. This unique mechanism of action leads to the occurrence of acute excitatory symptoms in pests and concurrently reduces the risk of cross-resistance development [7]. Beyond its application in rice cultivation, TFM has also shown excellent control efficacy against a broad spectrum of piercing-sucking pests in other major crops, including cotton, maize, and soybeans [8]. In 2017, the United States became the first country to grant approval for the use of TFM in rice pest control. Subsequently, TFM has obtained registration and approval for application in China, South Korea, India, Indonesia, Vietnam, and Japan [9]. It is anticipated that TFM will be registered in more countries in the near future, thereby enabling its global availability for agricultural pest management.

With the widespread application of TFM, its potential residues may pose risks to the safety of rice crops and the ecological environment of paddy fields. Existing studies have confirmed that even at the recommended field application doses, TFM exhibits significant acute toxicity; moreover, its sublethal concentrations can exert adverse effects on the growth, development, and biological functions of *Pterostichus fuscipes*, with the larval stage being particularly vulnerable [10]. Additionally, sublethal exposure to TFM can disrupt the enzymatic balance in key metabolic pathways of organisms, which may further impair their immune function [11,12]. Furthermore, relevant research has demonstrated that chronic exposure of fish to sublethal concentrations of TFM can induce harmful biochemical changes and histopathological damage [13]. To address the aforementioned risks and safeguard food safety, regulatory authorities across the globe have established Maximum Residue Limits (MRLs) for TFM in rice. Specifically, the Joint Meeting on Pesticide Residues (JMPR) in collaboration with the Food and Agriculture Organization (FAO) and the World Health Organization (WHO) recommends an MRL of 0.2 mg kg<sup>-1</sup> for TFM in rice [14]; Japan and the United States have set their respective MRLs at 0.01 mg kg<sup>-1</sup> and 0.4 mg kg<sup>-1</sup> [15,16]. At present, China has also formulated a provisional MRL for TFM in rice, which is 0.1 mg kg<sup>-1</sup>. Against this background, there is an urgent need to develop a diverse range of highly sensitive detection technologies for the accurate quantification of TFM residues. This effort serves as a crucial prerequisite for advancing the risk assessment of TFM and effectively implementing the established MRL requirements.

At present, the analytical methods reported for the determination of TFM residues mainly include high-performance liquid chromatography (HPLC) [17,18] and liquid chromatography-tandem mass spectrometry (LC-MS/MS) [[19], [20], [21], [22]]. However, HPLC exhibits low detection sensitivity and weak anti-interference ability when analyzing complex matrices. Although LC-MS/MS possesses excellent detection performance-for instance, the method established by Shen et al. (2022) for TFM detection in rice achieves a limit of quantitation (LOQ) of 0.03 mg kg<sup>-1</sup> [22]-this technique is associated with high costs for instrument purchase and maintenance, as well as the requirement for professional training to master operational skills. Given these limitations, many grassroots laboratories, facing strong demand for the detection of organophosphorus and organochlorine pesticides, tend to prioritize the deployment of gas chromatography (GC) instruments. Thus, the method for determining TFM residues in paddy field systems using GC technology possesses general applicability. GC boasts outstanding separation efficiency, balanced cost-effectiveness, and flexible compatibility with a variety of selective detectors [23,24]. More importantly, GC technology exhibits good anti-interference adaptability to a large number of non-polar impurities (such as lipids and pigments) present in paddy field matrices (especially rice husk and straw). HPLC/LC-MS/MS relies on polar mobile phases,

making it susceptible to interference from such non-polar impurities and thus requiring complex sample pretreatment processes, which significantly prolong the overall analysis cycle [25,26]. From the perspective of detection principle compatibility, GC-ECD demonstrates high sensitivity to compounds containing electronegative atoms. Notably, the TFM molecule contains multiple fluorine (F) atoms, whose structural characteristics are highly matched with the response characteristics of GC-ECD [26]. GC-MS, on the other hand, can achieve accurate separation of structurally similar compounds while ensuring rapid and high-sensitivity detection of complex mixtures [27]. Overall, GC-ECD/GC-MS techniques offer high sensitivity, cost-effectiveness, and well-established operational procedures, rendering them well-suited for large-scale TFM residue screening in grassroots paddy field scenarios, yet to date no studies have been reported on their use for TFM residue analysis across multiple paddy matrices.

Therefore, the primary objective of this study was to develop accurate, reliable, and sensitive GC-ECD and GC-MS methods for the determination of TFM residues in five key paddy matrices: paddy water, paddy soil, brown rice, rice husk, and rice straw. To the best of our knowledge, this is the first study to establish GC-based analytical methods for TFM and validate their applicability across rice matrices. The specific goals were to: (1) optimize the sample pretreatment process (extraction and purification) to eliminate matrix interference; (2) validate the methods in terms of linearity, limit of detection (LOD), LOQ, recovery, precision, and robustness; and (3) apply the validated methods to real field samples. The analytical methods established in this study can provide technical support for the registration of TFM and the routine monitoring of TFM residues in rice products. Meanwhile, these methods are expected to reduce detection costs and improve analysis efficiency for quality and safety supervision programs of agricultural products, and provide a theoretical basis for the accurate determination of TFM residues in the rice production chain.

## Section snippets

### Chemicals and reagents

TFM (CAS No 1263133-33-0) standards (with a purity of 99.5 %) were obtained from DuPont Company (Delaware, USA). HPLC-grade acetone, analytical grade acetonitrile, and dichloromethane were provided by Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China), while analytical-grade sodium chloride (NaCl) was procured from Yongda Chemical Reagent Co., Ltd. (Tianjin, China). Primary-secondary amine (PSA, 40  $\mu$ m), octadecylsilane (C18, 40  $\mu$ m), florisil (Flo, 40  $\mu$ m), and 0.22  $\mu$ m nylon syringe filters

### Optimizing the GC analytical conditions

The effect of inlet temperature on triflumezopyrim (TFM) response was investigated across 11 temperature levels (200–300 °C). For gas chromatography-electron capture detector (GC-ECD, Fig. 1A), the TFM response increased significantly with temperature in the range of 200–260 °C and plateaued at higher temperatures. An analogous trend was observed for gas chromatography-mass spectrometry (GC-MS, Fig. 1B), where the response rose sharply between 200 and 250 °C and stabilized thereafter. Inlet

### Summary and perspectives

This study established a novel method for the determination of TFM residues in multiple paddy field matrices, including paddy water, paddy soil, brown rice, rice husk, and rice straw. Key improvements were achieved through three targeted optimizations: (1) systematic screening of extraction solvents (dichloromethane, acetonitrile, and acetone); (2) matrix-specific selection of adsorbents (PSA, C18, Florisil, and their various combinations); and (3) refinement of instrumental parameters (inlet

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### CRedit authorship contribution statement

**Xinhong Deng:** Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Wenxing Wang:** Validation, Data curation. **Bin Yang:** Validation, Data curation. **Xuguo Zhou:** Supervision, Project administration. **Lihua Yang:** Writing – review & editing, Supervision, Project administration.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.